

1 Supplementary Study

The supplementary study investigated whether participants primarily focus on structural information or whether visual mechanism illustrations prompt them to form prior beliefs about both the underlying causal structure and the size of the structure’s parameters. To test this, we varied whether participants observed the full mechanism of the ball device or merely the lower section between the yellow and pink balls — the specific component conveying structural information (i.e., whether the machine is more likely to be a “causal” or a “non-causal” device).

An illustration of the stimuli presented across the different conditions is provided in Fig. 1. In light of the complications encountered with the adapted stimuli in Experiment 4, we reverted to our original machine illustrations. This study is a close replication of Experiment 4, with the sole exception that visual mechanism illustrations were utilized across all conditions. Like our other experiments, this study was pre-registered on the Open Science Framework (OSF). It can be accessed here: https://osf.io/utp7m/overview?view_only=35656dbefe62429baaae6496675b8de3.

1.1 Methods

1.1.1 Participants

Five hundred and thirty participants were recruited via Prolific (www.prolific.com). Eligible participants were required to be at least 18 years old, native English speakers, have an approval rate of at least 90%, participate via a PC or laptop, have no history of participation in the pilot studies, and possess at least some formal educational qualification (i.e., individuals reporting “no formal qualifications” were excluded).

One participant was excluded for failing to confirm participation via a PC or laptop. An additional 49 participants (9.26%) were excluded for failing to meet the prespecified learning-phase performance criterion, which was identical to the one used in Experiment 4. The final analysis sample consisted of $N = 480$ participants (223 male, 249 female, 7 non-binary, 1 preferred not to say; $M_{\text{age}} = 35$ years, range: 18–83 years). Data collection was terminated once each condition reached $n = 60$ participants post-exclusion, matching the sample size per condition established in our previous experiments.

1.1.2 Design, materials, and procedure

The study employed a 2 (amount of device mechanism information: partial vs. full; between-subjects) $\times 2$ (information type: suggesting S_0 vs. suggesting S_1 ; between-subjects) $\times 2$ (contingency: (4/4)(4/4) vs. (2/4)(2/4); between-subjects) $\times 2$ (causal rating type: prior vs. posterior; within-subjects) mixed design.

The experimental scenario and procedure identical to our previous experiments. Following general information about the ball device, participants advanced to a screen revealing its internal mechanism. The device illustration displayed on this screen varied by experimental condition (see Fig. 1). Participants were explicitly informed that this screen constituted their sole opportunity to study the device’s internal mechanism.

On the subsequent screen, participants viewed an illustration of the same device with its mechanism covered, and were asked to provide their prior causal structure rating. This

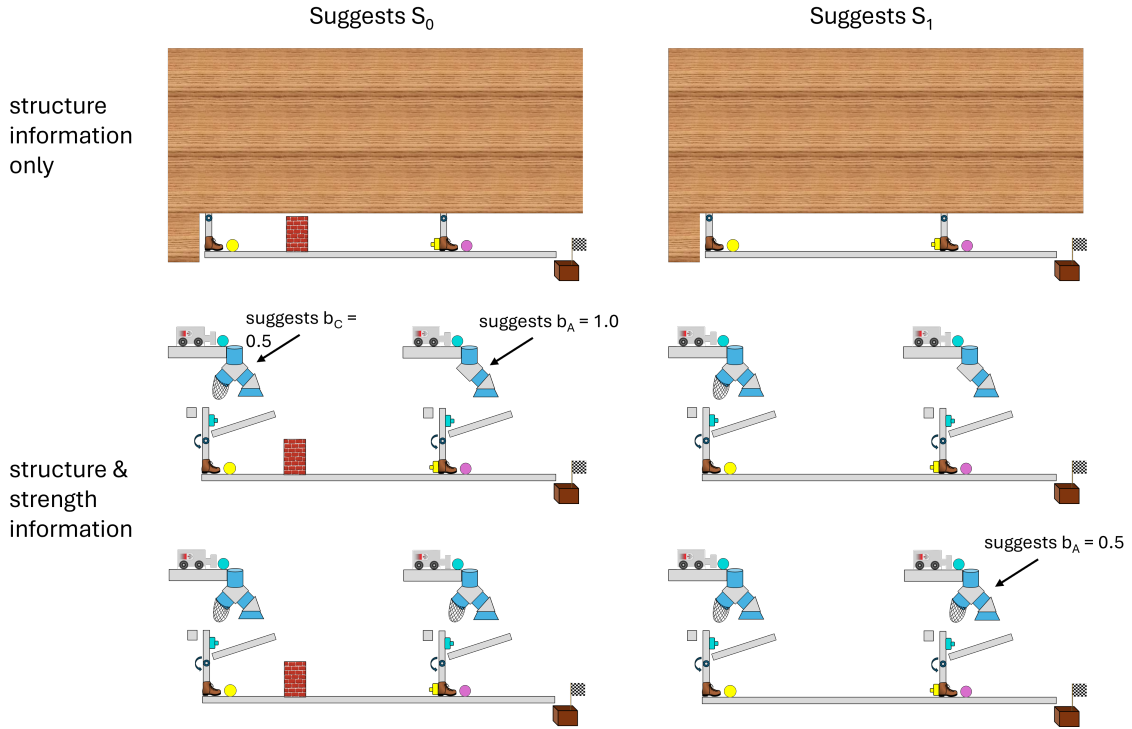


Figure 1

Illustration of the mechanism stimuli shown to participants in the supplementary study. The top two panels display the stimuli for the “partial mechanism information” condition. The remaining panels display the stimuli for the “full mechanism information” conditions, which varied based on the component representing the alternative cause.

query was the confidence query, like the one used in Experiment 3.

Participants then proceeded to the learning phase, which consisted of eight trials instantiating the respective contingency condition. Following the learning phase, participants returned to the rating screen to re-evaluate their confidence regarding whether the specific device was a “causal” one. Finally, participants provided demographic information, reported any technical issues, and completed the study via a brief debriefing screen.

1.2 Results and discussion

As illustrated in Fig. 2, the prior manipulation proved effective regardless of whether participants viewed the full or partial mechanism: participants who viewed illustrations without the wall (suggesting S_1) reported higher prior causal ratings than those who viewed illustrations featuring the wall (suggesting S_0). Furthermore, this group difference persisted in participants’ posterior causal ratings, with higher posterior ratings in the conditions suggesting S_1 than in those suggesting S_0 . While previous frameworks might interpret this finding as evidence of a “causal illusion,” the Bayesian learning perspective views this effect as a consequence of data providing insufficient counter-evidence against an ini-

tially high belief in S_1 .

Crucially, participants' rating behavior was similar across both levels of the device mechanism information factor. Whether participants observed the full mechanism or only the component conveying structural information had no impact on performance.

This descriptive pattern was supported by statistical analysis. A mixed-effects ANOVA—with amount of device mechanism information (ADMI), information type (IT; suggesting S_0 vs. S_1), contingency (C), and causal rating type (CRT) as fixed effects, and participant ID as a random effect—revealed neither a main effect of ADMI nor any significant interactions involving this factor.

However, there was a significant main effect of IT, $F(1, 303) = 1660.88$, $p < .001$, $\eta_p^2 = .846$, confirming that ratings were systematically higher when the displayed mechanism implied a causal link. A significant interaction also emerged between IT and CRT, $F(1, 303) = 102.37$, $p < .001$, $\eta_p^2 = .253$, indicating that the direction of rating shifts differed based on initial beliefs. As shown by the mean patterns in Fig. 2B, participants in the S_0 -suggesting conditions tended to slightly increase their ratings, whereas participants with a high prior belief in S_1 tended to decrease their ratings. Lastly, we observed a significant three-way interaction between IT, CRT, and C, $F(1, 303) = 3.90$, $p = .049$, $\eta_p^2 = .013$, demonstrating that the direction of these rating adjustments was also modulated by the specific learning data observed.

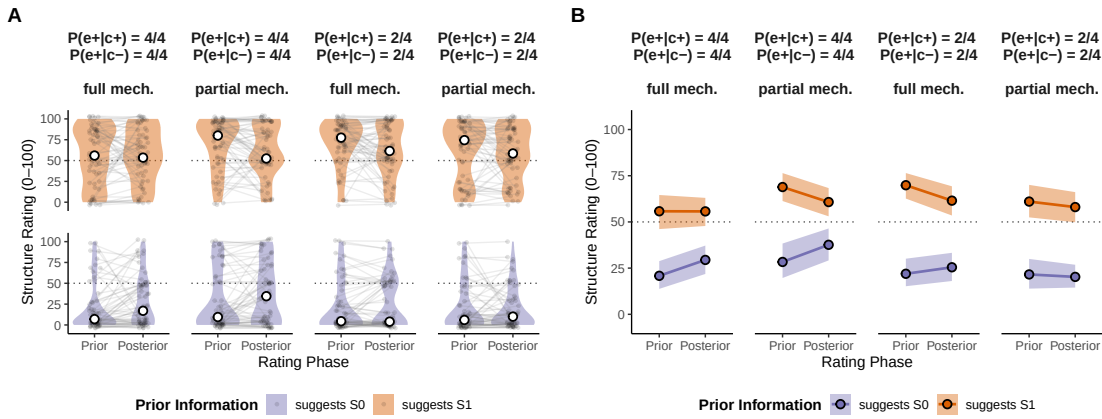


Figure 2

Results of the supplementary study. (A) Participants' prior and posterior causal structure ratings (jittered), with violin plots illustrating the distribution and white dots indicating medians. (B) Mean causal structure ratings, where ribbons represent 95% confidence intervals.

In sum, the supplementary study successfully replicates the influence of prior causal beliefs on subsequent causal judgments. Furthermore, these findings suggest that learners are primarily sensitive to structural information, whereas mechanism information detailing the strength of alternative causal variables does not exert a noticeable influence.